

Sampling Theorem & Discrete-Time Signals

Subject: Digital Signal Processing (EC3501) | Unit 1

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CORE CONCEPTS & LECTURE SUMMARY:

- Nyquist-Shannon Sampling Theorem criteria: $F_s \geq 2 \cdot F_{\max}$.
- Aliasing distortion and anti-aliasing low pass filter design specifications.
- Discrete Fourier Transform (DFT) mathematical formulation and twiddle factors.
- Z-Transform ROC (Region of Convergence) properties for stable, causal LTI systems.